Executive Business Report

Sales Forecasting & Demand Analytics for Retail Business

Prepared By: Pavan Kumar Verma
Project: Sales Forecasting & Demand Analytics Dashboard
Tools & Technologies: Python, Streamlit, Facebook Prophet, SARIMA, XGBoost, Isolation Forest, K-Means Clustering

GITHUB LINK : https://github.com/PavanvermaAKSU/End-to-End-Sales-Forecasting-intelligent-demand

STREAMLIT LINK: **https://salesforecastdashboard.streamlit.app**

Executive Summary

This project was conducted to analyze historical retail sales data and develop a data-driven decision support system for forecasting future sales, detecting unusual sales patterns, and improving inventory planning. Historical sales were analyzed using exploratory data analysis, followed by the implementation of three forecasting models: SARIMA, Facebook Prophet, and XGBoost. After evaluating each model using standard forecasting metrics, Facebook Prophet demonstrated the highest predictive accuracy and was selected as the recommended forecasting model. Additionally, anomaly detection techniques identified unusual sales events, while product demand segmentation grouped products with similar demand characteristics to support inventory optimization. The resulting dashboard enables business users to monitor sales performance, forecast future demand, and make informed operational decisions.

Key Findings from Exploratory Data Analysis

The exploratory analysis revealed several important business insights regarding historical sales performance:

- Sales demonstrated an overall upward trend between 2015 and 2018, indicating consistent business growth.
- Monthly sales exhibited noticeable seasonal patterns, suggesting that customer demand varies throughout the year.
- Technology and Furniture contributed a significant proportion of total sales revenue, making them key business categories.
- Regional sales performance varied across different geographical locations, indicating opportunities for region-specific inventory planning.
- Monthly sales fluctuations highlighted the need for an accurate forecasting system to improve inventory management and demand planning.

These findings provided the foundation for selecting an appropriate forecasting approach.

Three-Month Sales Forecast

Three forecasting techniques were evaluated, namely SARIMA, Facebook Prophet, and XGBoost. Based on model evaluation metrics, Facebook Prophet achieved the lowest forecasting error and was selected as the preferred production model.

FORECASTED MONTHLY SALES

Forecast Month	Forcasted Sales
Month 1	50900.79
Month 2	89710.82
Month 3	89211.29

The forecast indicates relatively stable sales over the next three months with sustained customer demand. Each prediction is accompanied by confidence intervals, representing the expected range within which actual sales are likely to occur under normal business conditions. These forecasts can assist management in procurement planning, budgeting, and inventory allocation.

Top Three Sales Anomalies

Anomaly detection was performed using Isolation Forest and validated using a rolling Z-Score approach. Several unusual sales observations were identified.

1. Significant Sales Spike

Likely Business Explanation

- A sudden increase in sales is likely associated with promotional campaigns, festive shopping periods, or large-volume customer purchases.

2. Unexpected Sales Decline

Likely Business Explanation

- A sharp decrease in sales may indicate inventory shortages, supply chain disruptions, reduced customer demand, or operational constraints.

3. Exceptional Peak Sales Event

Likely Business Explanation

- An unusually high sales value may correspond to end-of-season clearance sales, special discount events, or bulk corporate orders.
- Monitoring such anomalies enables the business to understand exceptional events and improve future operational planning.

PRODUCT DEMAND SEGMENTATION

DEMAND SEGMENT	BUSINESS CHARACTERSTICS	RECOMMENDED STARATEGY
High Volume, stable demand	Consistently strong demand	Maintain high inventory levels with frequent replenishiment
High Value, High Volatility	High revenue but unpredictable demand	Maintain safety stock and monitor demand closely

Growing demand	Increasing sales trend	Gradually increase inventory to accommodate future growth
Low Volume, Stable Demand	Predictable but relativity low demand	Maintain minimum stock to reduce inventory holldings costs

Business Recommendations

1. Deploy Facebook Prophet for Operational Forecasting

Facebook Prophet consistently outperformed SARIMA and XGBoost across all evaluation metrics.

Supporting Evidence

- Lowest MAE: **9,839.84**
- Lowest RMSE: **14,133.08**
- Lowest MAPE: **15.67%**

Its forecasting accuracy makes it suitable for monthly demand planning and inventory forecasting.

2. Implement Demand-Based Inventory Planning

Inventory policies should be aligned with product demand segments. High-demand products should receive procurement priority, while volatile products should be monitored with dynamic safety stock policies.

This approach can improve product availability while reducing excess inventory costs.

3. Integrate Continuous Anomaly Monitoring

The anomaly detection system should be incorporated into routine sales monitoring. Early identification of unusual sales spikes or declines enables management to respond proactively through pricing adjustments, inventory redistribution, or targeted marketing campaigns.

Risk and Limitation

The forecasting system relies primarily on historical sales data. Consequently, unexpected external factors such as economic fluctuations, competitor actions, policy changes, supply chain disruptions, or unforeseen market events may affect future sales and reduce forecasting accuracy. Regular model retraining with updated sales data is recommended to maintain forecasting performance.

Conclusion

This project demonstrates the practical application of machine learning and time-series forecasting techniques for retail sales analytics. By integrating forecasting, anomaly detection, and demand segmentation into an interactive Streamlit dashboard, the system provides valuable decision support for supply chain management, inventory planning, and business forecasting. Based on quantitative evaluation, Facebook Prophet is recommended as the primary forecasting model due to its superior predictive performance and suitability for business applications. The developed solution offers a scalable framework that can support future enhancements, including real-time forecasting, automated inventory optimization, and advanced business intelligence reporting.